R-Package ‘MrImputation II’

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Type Package
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Title Multiple Ratio Imputation II
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Depends `R(>=4.5.0)`
Description `R`-Package `MrImputation II` implements the methods proposed by Takahashi (2025) and Takahashi (2026). This is an extended version of Takahashi (2017a, 2017b), which originally implemented `R`-Package `MrImputation`.
URL https://github.com/mtakahashi123/MrImputationII
Repository GitHub

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`MrImputation-package` *Multiple ratio imputation with diagnostic tests*

Description

To use this package, click "Code" and "Download ZIP" at https://github.com/mtakahashi123/MrImputationII. After downloading the package, set the working directory in `R`, and read `R`-Package `MrImputationII` using `R`-function `source` by either of the following methods.

Method 1

```
setwd("C:/Folder")
source("MrImputationII.R")
```

Method 2

```
source(file.choose())
```

References

Takahashi, M. (2026) "Adaptive Multiple Imputation for Heteroskedastic Economic Data: A Data-Driven Flexible Ratio Estimator Approach."

Takahashi, M. (2025) "The treatment of missing values in official statistics." *The Journal of Economics*, 65 (5/6), 125-136.

Takahashi, M. (2017a) "Multiple ratio imputation by the EMB algorithm: Theory and simulation." *Journal of Modern Applied Statistical Methods*, 16 (1), 630-656.

Takahashi, M. (2017b) "JMASM44: Implementing multiple ratio imputation by the EMB Algorithm (R)." *Journal of Modern Applied Statistical Methods*, 16 (1), 657-673.

`BPTtest` *R-function to diagnose the estimated theta*

Description

`BPTtest` implements the Breusch-Pagan-Takahashi test for the degree of heteroskedasticity.

Usage

```
BPTtest(y, x, theta = 0)
```

Arguments

`y`	Target variable for imputation
`x`	Auxiliary variable for the imputation model
`theta`	Value for the null hypothesis, i.e., the degree of heteroskedasticity. Default is 0.

Value

`BPT`	LM statistic for the BPT test for heteroskedasticity
`df`	Degrees of freedom
`p.value`	P-value. If the significance level is 0.05 and the p-value is less than 0.05, we reject the null hypothesis at the 5% error level, meaning that the estimated theta is considered to be wrong.

References

Takahashi, M. (2026) "Adaptive Multiple Imputation for Heteroskedastic Economic Data: A Data-Driven Flexible Ratio Estimator Approach."

Takahashi, M. (2025) "The treatment of missing values in official statistics." *The Journal of Economics*, 65 (5/6), 125-136.

Breusch, T. S. and Pagan, A. R. (1979) "A simple test for heteroscedasticity and random coefficient variation." *Econometrica*, 47 (5), 1287-1294.

Example

```
BPTtest(var1, var2, theta = 1.8956)
```

`ExampleData00.csv` *Simulated data for example 1*

Description

This is a simulated dataset as an example. The true value of theta is set to 0.0.

Usage

```
data1 <- read.csv("ExampleData00.csv", header=TRUE)
attach(data1)
```

Format

A data frame with 1000 observations on the following two variables.

`var1` the target variable for imputation with 285 observations missing.

`var2` the auxiliary variable for the imputation model.

`ExampleData05.csv` *Simulated data for example 2*

Description

This is a simulated dataset as an example. The true value of theta is set to 0.5.

Usage

```
data1 <- read.csv("ExampleData05.csv", header=TRUE)
attach(data1)
```

Format

A data frame with 1000 observations on the following two variables.

`var1` the target variable for imputation with 285 observations missing.

`var2` the auxiliary variable for the imputation model.

`ExampleData10.csv` *Simulated data for example 3*

Description

This is a simulated dataset as an example. The true value of theta is set to 1.0.

Usage

```
data1 <- read.csv("ExampleData10.csv", header=TRUE)
attach(data1)
```

Format

A data frame with 1000 observations on the following two variables.

`var1` the target variable for imputation with 285 observations missing.

`var2` the auxiliary variable for the imputation model.

`mranalyze2` *R-function to analyze multiply-imputed datasets after multiple ratio imputation*

Description

`mranalyze2` computes the mean, the standard error, the confidence intervals, the correlation coefficient, and the degrees of freedom after multiple ratio imputation. It automatically combines *M* estimates.

Usage

```
mranalyze2(data, alpha = 0.05)
```

Arguments

`data`	Multiply-imputed data after `mrimpute2`
`alpha`	Significance level. Default is 0.05.

Value

`ybar`	Sample mean of the target variable
`se`	Standard error of the sample mean
`CI.LL`	Lower limit of the confidence interval
`CI.UL`	Upper limit of the confidence interval
`corr`	Correlation coefficient
`df`	Degrees of freedom

References

Takahashi, M. (2026) "Adaptive Multiple Imputation for Heteroskedastic Economic Data: A Data-Driven Flexible Ratio Estimator Approach."

Takahashi, M. (2017a) "Multiple ratio imputation by the EMB algorithm: Theory and simulation." *Journal of Modern Applied Statistical Methods*, 16 (1), 630-656.

Takahashi, M. (2017b) "JMASM44: Implementing multiple ratio imputation by the

EMB Algorithm (R)." *Journal of Modern Applied Statistical Methods*, 16 (1), 657-673.

Example

```
data1 <- read.csv("ExampleData05.csv", header=TRUE)
attach(data1)
imp1 <- mrimpute2(var1, var2)
mranalyze2(imp1)
mranalyze2(imp1, alpha = 0.01)
```

`mrdiag` *R-function to produce diagnostic plot for multiply-imputed datasets after multiple ratio imputation*

Description

`mrdiag` plots the density of observed data and multiply-imputed data.

Usage

```
mrdiag(data)
```

Arguments

`data`	Multiply-imputed data after `mrimpute2`

References

van Buuren, S. (2018) *Flexible Imputation of Missing Data*, 2nd ed. CRC Press.

Example

```
data1 <- read.csv("ExampleData05.csv", header=TRUE)
attach(data1)
imp1 <- mrimpute2(var1, var2)
mrdiag(imp1)
```

`mrimpute2` *R-function to compute multiple ratio imputation*

Description

`mrimpute2` multiply-imputes missing values based on multiple ratio imputation.

Usage

```
mrimpute2(y, x, M = 5, iter = 5)
```

Arguments

`y`	Target variable for imputation
`x`	Auxiliary variable for the imputation model
`M`	Number of multiply-imputed datasets. Default is 5.
`iter`	Number of iterations to estimate the degree of heteroskedasticity. Default is 5.

References

Takahashi, M. (2026) "Adaptive Multiple Imputation for Heteroskedastic Economic Data: A Data-Driven Flexible Ratio Estimator Approach."

Takahashi, M. (2017a) "Multiple ratio imputation by the EMB algorithm: Theory and simulation." *Journal of Modern Applied Statistical Methods*, 16 (1), 630-656.

Takahashi, M. (2017b) "JMASM44: Implementing multiple ratio imputation by the EMB Algorithm (R)." *Journal of Modern Applied Statistical Methods*, 16 (1), 657-673.

Example

```
data1 <- read.csv("ExampleData05.csv", header=TRUE)
attach(data1)
imp1 <- mrimpute2(var1, var2)
imp2 <- mrimpute2(var1, var2, M = 100)
imp3 <- mrimpute2(var1, var2, M = 100, iter = 10)
```

Value

A data frame with n observations on the following $M + 2$ variables.

`V1` the target variable for imputation. The same as `var1`.

`V2` the auxiliary variable for the imputation model. The same as `var2`.

`V3` multiply-imputed data of `var1`, where $m = 1$.

`V4` multiply-imputed data of `var1`, where $m = 2$.

⋮

`VM+2` multiply-imputed data of `var1`, where $m = M$.

`outputimp.csv` *Multiply-imputed data for example*

Description

This is an example of multiply-imputed dataset after multiple ratio imputation by `R` function `mrimpute2`.

Usage

```
outputimp <- read.csv("outputimp.csv", header=TRUE)
attach(outputimp)
```

Format

A data frame with 1000 observations on the following six variables.

`V1` the target variable for imputation. The same as `var1`.

`V2` the auxiliary variable for the imputation model. The same as `var2`.

`V3` multiply-imputed data of `var1`, where $m = 1$.

`V4` multiply-imputed data of `var1`, where $m = 2$.

`V5` multiply-imputed data of `var1`, where $m = 3$.

`V6` multiply-imputed data of `var1`, where $m = 4$.

`V7` multiply-imputed data of `var1`, where $m = 5$.

`thetahat` *R-function to estimate the degree of heteroskedasticity*

Description

`thetahat` estimates the degree of heteroskedasticity for ratio imputation.

Usage

```
thetahat(y, x, iter = 5, convplot = TRUE)
```

Arguments

`y`	Target variable for imputation
`x`	Auxiliary variable for the imputation model
`iter`	Number of iterations to estimate the degree of heteroskedasticity. Default is 5.
`convplot`	Convergence plot to check whether the estimated theta converges to a certain point. Default is TRUE.

Value

`theta.hat`	Estimated value of theta, which is the degree of heteroskedasticity.

References

Takahashi, M. (2026) "Adaptive Multiple Imputation for Heteroskedastic Economic Data: A Data-Driven Flexible Ratio Estimator Approach."

Example

```
t1 <- thetahat(var1, var2)
t1 <- thetahat(var1, var2, iter = 100)
t1 <- thetahat(var1, var2, iter = 100, convplot = FALSE)
```